Autonomous Driving Project

AI for Automotive
Electronic Engineering for Intelligent Vehicles

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Abstract

This project presents an end-to-end autonomous driving model for predicting continuous vehicle control commands, specifically steering angle and acceleration, from raw video data and vehicle state information. Inspired by prior work on end-to-end driving models, the proposed system leverages a Vision Transformer for visual feature extraction and a Transformer-based temporal module for modeling driving dynamics over short time horizons. The model is trained and evaluated on the Berkeley DeepDrive Video (BDDV) dataset using a regression-based formulation. Experimental results on held-out driving sequences show that the proposed architecture produces smooth and temporally consistent control predictions, supporting the effectiveness of Transformer-based models for imitation learning in autonomous driving.

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1 Introduction

Autonomous driving requires learning a mapping from high-dimensional sensory inputs, such as video streams and vehicle state measurements, to continuous control commands in real time. Traditional approaches typically rely on modular pipelines that separate perception, planning, and control. While effective, these systems require extensive manual design and may suffer from error propagation across components.

End-to-end learning provides an alternative paradigm by directly predicting driving commands from raw sensory data. In this work, we adopt an end-to-end imitation learning approach for continuous steering angle and acceleration prediction, leveraging modern Transformer-based architectures to model both visual information and temporal driving dynamics.

The proposed model is trained and evaluated on the Berkeley DeepDrive Video (BDDV) dataset, which contains front-facing dashcam videos synchronized with vehicle sensor data collected in real-world driving conditions. Each input sample consists of a short temporal window of consecutive frames and corresponding sensor measurements, with the objective of predicting continuous control commands for the next time step.

The goal of this project is to evaluate whether a Transformer-based visual encoder, combined with a Transformer-based temporal fusion module, can effectively learn driving behavior from video data and produce smooth and consistent control predictions within a realistic driving dataset.

2 Method

In this project, we developed an *end-to-end* deep learning model aimed at predicting continuous driving commands, specifically the steering angle and acceleration, directly from raw video frames and sensor data. The core idea behind the approach is to combine *spatial feature extraction* from individual frames with *temporal modeling* to capture the dynamics of driving behavior over time.

2.1 Network Architecture

The proposed architecture follows an end-to-end imitation learning paradigm and is composed of three main components: a visual encoder, a temporal fusion module, and a regression head (Figure 1).

Visual Encoder: For visual feature extraction, a pre-trained Vision Transformer (ViT) is used as a fixed backbone. Each input frame is resized, divided into patches, and processed independently by the ViT to produce a compact, high-level visual representation. The ViT enables the modeling of global spatial context and long-range dependencies, which are important for understanding complex driving scenes such as intersections and curved roads. The output representation is given by the class (CLS) token for each frame.

Temporal Fusion Module: To model temporal dependencies and vehicle motion, the sequence of frame embeddings (CLS tokens) is concatenated with the corresponding vehicle sensor data and passed to a Transformer-based temporal encoder. Each embedding is treated as a state token, and temporal self-attention is used to capture motion patterns, vehicle dynamics, and correlations between consecutive observations. This temporal modeling allows the network to reason over short driving histories instead of relying on single-frame predictions, which is essential for smooth and stable control.

Regression Head: The output of the temporal Transformer is fed into a lightweight regression head implemented as a multilayer perceptron (MLP). The regression head performs a multi-output regression task and predicts two continuous control variables: steering angle and acceleration magnitude.

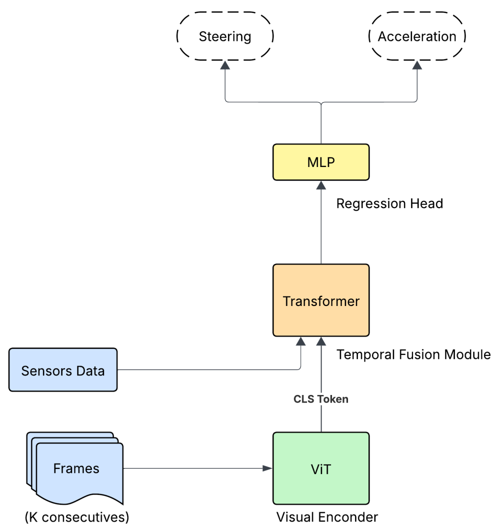

Figure 1: Architecture of the proposed end-to-end driving model. Visual features extracted by a frozen Vision Transformer and vehicle sensor data are fused by a Transformer-based temporal module. A regression head predicts continuous steering and acceleration commands.

2.2 Loss Function

The model is trained using supervised regression losses computed independently for each control variable. The total training loss is defined as:

$$J(\theta) = \lambda_s L_{steer} + \lambda_a L_{accel} \tag{1}$$

where L_{steer} and L_{accel} denote the Mean Squared Error (MSE) losses for steering and acceleration, respectively, and λ_s and λ_a are weighting coefficients. The MSE loss is defined as:

$$\frac{1}{n}\sum_{i=1}^{n}(\hat{y}^{(i)} - y^{(i)})^2 \tag{2}$$

A higher weight was assigned to the steering loss to reflect the greater complexity and safety relevance of steering prediction compared to acceleration. Without this weighting, the model tended to bias toward trivial straight-driving solutions due to the dominance of low-curvature segments in the dataset.

Additionally, steering events occur for a significantly shorter duration relative to the overall training video length, leading to an inherent class imbalance between straight and turning maneuvers. By increasing the steering loss weight, the model is encouraged to learn meaningful steering dynamics rather than minimizing loss through near-zero predictions.
This adjustment resulted in a noticeable improvement in steering behavior, particularly after the second training phase, where the model demonstrated enhanced responsiveness and better alignment with ground-truth steering during turning events.

2.3 Training Strategy

Different parts of the network are trained using different strategies to ensure stable learning given the limited dataset size:
Vision Transformer

- Initialized with pre-trained weights
- Kept frozen during training
- Used as a fixed visual feature extractor

Temporal Transformer

- Trained from scratch
- Learns driving-specific temporal dynamics

Regression Head

- Trained from scratch
- Maps temporal features to continuous control outputs

This training strategy uses strong pre-trained visual representations while allowing task-specific temporal and control learning, leading to stable and effective end-to-end training.

3 Design Considerations

Several design choices were made to ensure the model is suitable for real-world driving scenarios:

- **Sequence length:** A fixed number of consecutive frames is used to balance temporal context and computational efficiency.
- **Normalization:** Input frames and sensor values are normalized to improve numerical stability and convergence during training.
- **Regularization:** Dropout is applied in the regression head to reduce overfitting and improve generalization.

Overall, this architecture enables the model to understand the current driving scene, model vehicle dynamics over short temporal horizons, and output accurate continuous driving commands in an end-to-end manner.

4 Implementation Details

This section describes the practical implementation details of the proposed system, including data preprocessing, training hyperparameters, model initialization choices, and the software environment. These details complement the methodological description in Chapter 2 and are provided to ensure reproducibility and consistency with the implemented codebase.

The model was implemented in **PyTorch** and trained on the university computing cluster. A subset of approximately 1,000 video clips from the BDDV dataset was used for training, while the full validation and test splits were used for evaluation.

5 Data Preprocessing

Video frames were resized to 224×224 pixels to match the input resolution of the Vision Transformer and normalized using the standard ImageNet mean and variance. Each training sample consists of a temporal window of $K = 10$ consecutive frames extracted from the same video sequence, ensuring correct temporal ordering.

Steering angle and acceleration signals were normalized prior to training, and extreme outliers were clipped to improve numerical stability and prevent gradient instability. To mitigate the inherent dataset bias toward straight driving, the training subset was curated to ensure adequate representation of curved road segments, intersections, and other non-straight driving maneuvers. For the steering action, yaw-rate measurements from the gyroscope were used as a reference signal and correlated with the corresponding video frames to derive a consistent steering representation. This fusion of visual data with inertial yaw information enabled more reliable supervision of steering behavior, particularly during short-duration turning events.

6 Training Parameters

The network was trained using the **Adam** optimizer with a learning rate of 1×10^{-4}. Due to GPU memory constraints and the use of long video sequences, a batch size of 2 was employed. Training was performed for 5 epochs.

The total training objective is a weighted Mean Squared Error (MSE) loss over steering angle and acceleration, with a higher weight assigned to the steering term to encourage accurate learning of turning behavior. No explicit gradient clipping was applied during training.

Temporal windows were extracted using a stride equal to K, avoiding overlap between consecutive samples and reducing temporal correlation within training batches.

Table 1: Summary of training and model hyperparameters.

Hyperparameter	**Value**
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	2
Number of epochs	5
Temporal window size (K)	10 frames
Temporal stride	K (no overlap)
Loss function	Weighted MSE
Steering loss weight	$\lambda_s > \lambda_a$
Acceleration loss weight	λ_a
ViT backbone	Pre-trained, frozen
Temporal Transformer hidden size	256
MLP hidden units	128
MLP hidden layers	1
Activation function	ReLU
Dropout	Enabled (MLP)
Gradient clipping	Not used
Weight initialization	PyTorch default

7 Model Initialization

The Vision Transformer (ViT) was initialized with publicly available pre-trained weights and kept **fully frozen** during training, acting as a fixed visual feature extractor. This design choice follows best practices for transfer learning when the available task-specific data is limited.

The temporal Transformer encoder and the regression head were trained from scratch using PyTorch's default parameter initialization. The regression head is implemented as a multilayer perceptron (MLP) with a hidden dimension of 128, ReLU activation, and dropout regularization.

8 Results

This section presents the experimental results of the proposed end-to-end driving model, evaluated on the BDDV validation set. Both quantitative regression metrics and qualitative prediction plots are reported to assess the accuracy, stability, and temporal consistency of the predicted steering and acceleration commands.

9 Quantitative Results

The model was evaluated on 11,940 validation clips using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Squared Error (MSE). Metrics were computed independently for steering angle and acceleration. Table 2 summarizes the quantitative results.

Table 2: Regression performance of the proposed model on the BDDV validation set.

Output	**MAE**	**RMSE**
Steering angle (normalized)	0.0560	0.0969
Acceleration (normalized)	0.0447	0.0551

The overall Mean Squared Error (MSE) across both outputs is 0.0062. As expected, steering prediction exhibits higher error than acceleration due to its higher variability and stronger dependence on visual context, particularly in curved-road and turning scenarios.

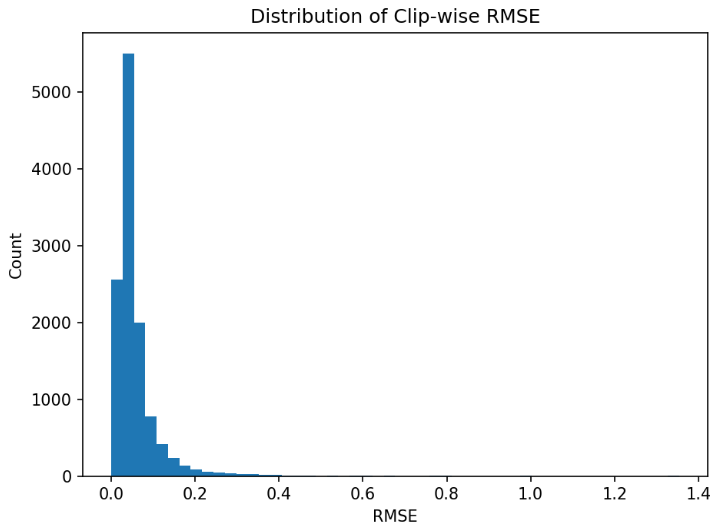

Figure 2: Distribution of clip-wise RMSE over the validation set.

Figure 2 shows the distribution of clip-wise RMSE values over the validation set. The distribution is strongly right-skewed, with the majority of clips exhibiting low prediction error and a small number of clips producing higher RMSE values. These higher-error cases typically correspond to more challenging driving scenarios, such as sharp turns or complex maneuvers. Overall, the histogram indicates that the model performs consistently across most clips, while a limited number of difficult sequences dominate the tail of the error distribution.

10 Qualitative Results

Figure 3 shows an example sequence comparing the predicted steering and acceleration values with the ground-truth signals over time. The predictions closely follow the overall trends of the true driving commands and exhibit smooth temporal behavior.

The qualitative results confirm that the temporal Transformer effectively enforces temporal consistency, reducing abrupt changes and producing smooth control signals suitable for driving applications.

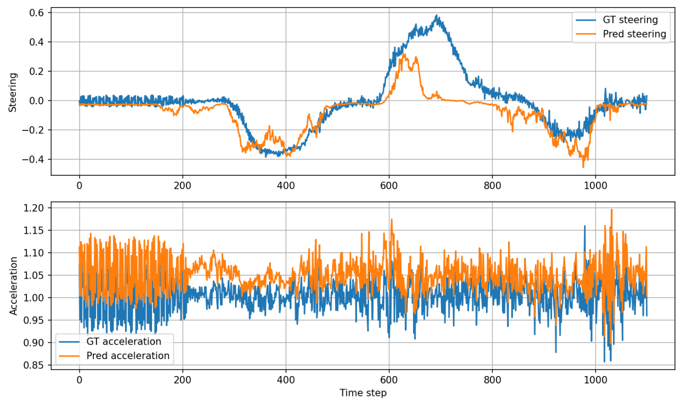

Figure 3: Example predictions vs. ground-truth for steering and acceleration over a video sequence.

Figure 4: Predictions and ground-truth overlaid on a video frame.

11 Discussion

The results demonstrate that the proposed Transformer-based architecture can accurately predict continuous driving commands from video data. Quantitative metrics indicate that acceleration is easier to predict than steering, reflecting the higher complexity and sensitivity of steering behavior. Qualitative analysis further confirms that the model generalizes across different driving situations and captures both straight-road and curved-road dynamics with stable temporal behavior.

12 Conclusions

This project investigated an end-to-end learning approach for autonomous driving, focusing on the prediction of continuous steering angle and acceleration commands from raw video frames and sensor data. The proposed model, based on a pre-trained Vision Transformer for spatial feature extraction and a Transformer-based temporal module for scene dynamics modeling, was successfully trained using imitation learning on the BDDV dataset. Quantitative results show that the model predicts acceleration more accurately than steering, reflecting the higher complexity and variability of steering behavior. Qualitative evaluations further demonstrate that the model produces smooth and temporally consistent predictions, capturing both straight-road and curved-road driving patterns across diverse scenarios within the dataset. Several design choices contributed to the observed performance:

- The Vision Transformer provides high-level spatial representations from raw video frames.
- The temporal Transformer effectively models short-term driving dynamics, improving temporal consistency.
- Loss reweighting, dataset balancing, and proper normalization were essential for stable training and accurate regression performance.

Despite these promising results, the model was evaluated only on held-out sequences from the same dataset and over short temporal horizons. Future work could explore longer temporal windows, data augmentation strategies, fine-tuning the visual encoder with additional data, or incorporating additional sensor modalities to further improve performance, particularly in complex driving scenarios such as sharp turns or dense traffic.